WORKED SOLUTION

STEP II 2005, Question 6

Pure • Binomial Series Sums

STEP II 2005, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q6

The key idea

Every sum here is a power series evaluated at a particular x, so the job is to recognise which function it comes from. In part (i) the trick is to rewrite n and n^2 as combinations of 1, $n+1$ and $(n+1)(n+2)$, which are exactly the coefficients we have just written down. In part (ii) the same idea works once we notice how the coefficients of $(1-x)^{-1/2}$ and $(1-x)^{-3/2}$ are related.

Solution

Part (i)

By the binomial theorem for negative integer powers (valid for $|x|<1$), the coefficients of x^n, for $n \geq 0$, are

$$(1-x)^{-1}: \quad 1, \qquad (1-x)^{-2}: \quad n+1, \qquad (1-x)^{-3}: \quad \tfrac{1}{2}(n+1)(n+2).$$

For instance, the coefficient in $(1-x)^{-3}$ is $\dfrac{(-3)(-4)\cdots(-3-n+1)}{n!}(-1)^n = \dfrac{3\cdot 4\cdots(n+2)}{n!} = \dfrac{(n+1)(n+2)}{2}$.

The first sum. Since $n = (n+1)-1$, for $|x|<1$

$$\sum_{n=0}^{\infty} nx^n = \sum_{n=0}^{\infty}(n+1)x^n - \sum_{n=0}^{\infty} x^n = \frac{1}{(1-x)^2} - \frac{1}{1-x}.$$

(Splitting the sum is fine because both series on the right converge.) The $n=0$ term is zero, so putting $x = \frac{1}{2}$,

$$\sum_{n=1}^{\infty} n\,2^{-n} = 4 - 2 = 2.$$

The second sum. We want n^2 as a combination of $(n+1)(n+2)$, $n+1$ and 1. Since

$(n+1)(n+2) = n^2 + 3n + 2$,

$$n^2 = (n+1)(n+2) - 3(n+1) + 1.$$

Hence, for $|x| < 1$,

$$\sum_{n=0}^{\infty} n^2 x^n = \frac{2}{(1-x)^3} - \frac{3}{(1-x)^2} + \frac{1}{1-x},$$

where the factor 2 appears because the coefficient in $(1-x)^{-3}$ is *half* of $(n+1)(n+2)$. Again the $n = 0$ term vanishes, and at $x = \frac{1}{2}$

$$\sum_{n=1}^{\infty} n^2 2^{-n} = 2 \cdot 8 - 3 \cdot 4 + 2 = 6.$$

Part (ii)

By the binomial theorem with index $-\frac{1}{2}$, the coefficient of x^n in $(1-x)^{-1/2}$ is

$$\frac{(-\frac{1}{2})(-\frac{3}{2}) \cdots (-\frac{2n-1}{2})}{n!} (-1)^n = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n \, n!}.$$

Ask yourself How do we turn a product of odd numbers into factorials?

Fill in the missing even numbers: $(2n)! = (1 \cdot 3 \cdots (2n-1))(2 \cdot 4 \cdots 2n)$, and $2 \cdot 4 \cdots 2n = 2^n n!$. So $1 \cdot 3 \cdots (2n-1) = \dfrac{(2n)!}{2^n n!}$, and the coefficient is

$$a_n = \frac{(2n)!}{2^n n! \cdot 2^n n!} = \frac{(2n)!}{(n!)^2} \frac{1}{2^{2n}}.$$

This holds for $n = 0$ too (empty product, $a_0 = 1$), so for $|x| < 1$

$$(1-x)^{-1/2} = \sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2} \frac{x^n}{2^{2n}},$$

as required.

The first sum. The sum asked for is $\sum a_n x^n$ at $x = \frac{1}{3}$, which is inside the interval of validity. So it equals

$$\left(1 - \tfrac{1}{3}\right)^{-1/2} = \sqrt{\tfrac{3}{2}} = \frac{\sqrt{6}}{2}.$$

The second sum. Now we need $\sum n a_n x^n$. In the spirit of part (i), relate na_n to a series we know. Exactly as above, the coefficient of x^m in $(1-x)^{-3/2}$ is

$$b_m = \frac{3 \cdot 5 \cdots (2m+1)}{2^m \, m!} = (2m+1)\, a_m.$$

From the formula for a_n, the ratio of consecutive coefficients is $a_n/a_{n-1} = \dfrac{(2n)(2n-1)}{n^2 \cdot 4} = \dfrac{2n-1}{2n}$, so for $n \geq 1$

$$n\,a_n = \tfrac{1}{2}(2n-1)\,a_{n-1} = \tfrac{1}{2}\,b_{n-1}.$$

Therefore, for $|x| < 1$,

$$\sum_{n=1}^{\infty} n a_n x^n = \frac{x}{2}\sum_{n=1}^{\infty} b_{n-1}x^{n-1} = \frac{x}{2}(1-x)^{-3/2}.$$

(Equivalently, differentiate the series for $(1-x)^{-1/2}$ term by term, which is allowed inside its interval of convergence, and multiply by x.) At $x = \frac{1}{3}$:

$$\sum_{n=1}^{\infty} \frac{n\,(2n)!}{(n!)^2\,2^{2n}\,3^n} = \frac{1}{6}\left(\frac{3}{2}\right)^{3/2} = \frac{1}{6}\cdot\frac{3}{2}\sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{8}.$$

Checks

- The first few terms of $\sum n2^{-n}$ are $\frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \frac{1}{4} + \cdots$, already 1.625 after four terms and creeping towards 2; partial sums of $\sum n^2 2^{-n}$ to 200 terms agree with 6 to many decimal places.
- Numerically, $\sqrt{6}/2 \approx 1.224745$ and $\sqrt{6}/8 \approx 0.306186$; the partial sums of the two series in part (ii) match both to 12 decimal places.
- A quick sanity check on the general term: $n = 1$ gives $a_1 = \frac{2}{4} = \frac{1}{2}$, the familiar coefficient in $(1-x)^{-1/2} = 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \cdots$.

Common slips

- Forgetting the $\frac{1}{2}$ in the coefficient of $(1-x)^{-3}$, which gives $8 - 12 + 2 = -2$ for $\sum n^2 2^{-n}$, impossible for a sum of positive terms.
- Starting the sum for $(1-x)^{-2}$ with coefficient n rather than $n+1$: the series is $1 + 2x + 3x^2 + \cdots$.
- In the last sum, losing a factor of x: $\sum n a_n x^{n-1}$ is the derivative, but the sum asked for is $\sum n a_n x^n$.

Beyond STEP

The manoeuvres here (splitting a series into pieces, differentiating term by term) are legitimate because a power series converges absolutely inside its radius of convergence, and can be differentiated there term by term. Proving that carefully is a standard part of a first course in analysis. The coefficients $\binom{2n}{n}/4^n$ also turn up in probability, as the

chance that a simple random walk is back at its start after $2n$ steps. For the analysis, see Abbott, *Understanding Analysis*.

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